Securing Routing Protocols:

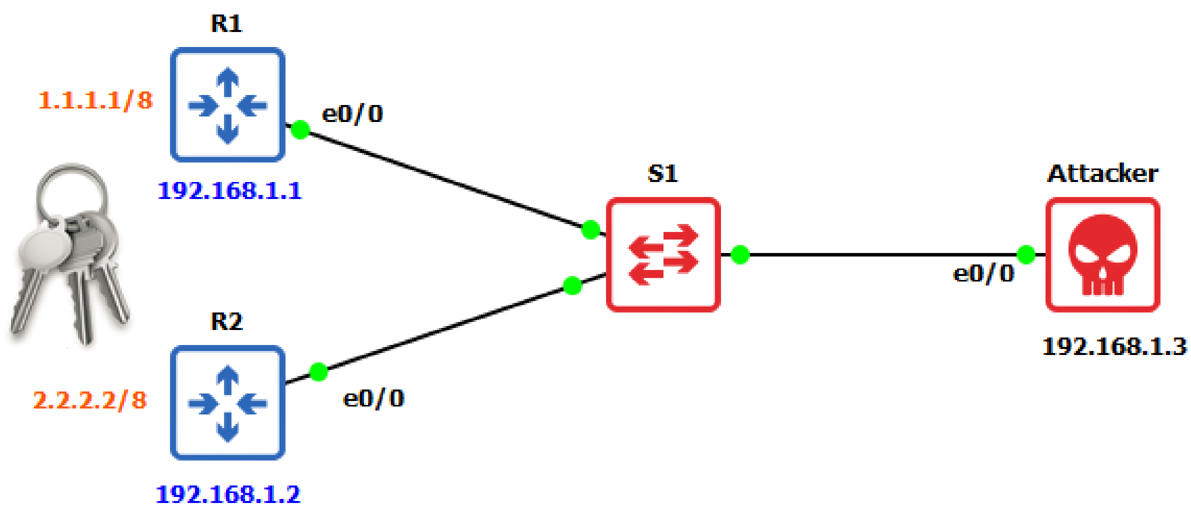
- By default, all, the routing information is visible to all interested parties.
- By default, routing information is not encrypted therefore open to an attack.
- Routing protocols can be configured to prevent receiving false routing updates.
- Routing Protocol Authentication is a security mechanism prevent such attacks.
- To prevent router from accepting unauthorized or malicious routing updates.
- Routing protocols RIP, EIGRP and OSPF can be secure by putting an authentication
- Two options for authentication are Plain text authentication & MD5 authentication.

Authentication on OSPF:

- OSPF Dynamic Routing protocols can authenticate every OSPF message to protect.
- This is basically to prevent a rogue Router from injecting false routing information.
- This is also to prevent the Denial-of-Service (DoS) attack on Cisco Devices as well.
- Two types of authentication can be used one of them is Plain text authentication.
- Second is MD5 authentication, MD5 type of Authentication is stronger and secure.
- There are all together three types of OSPF authentication Type 1, Type 2 and Type 3.
- Type 1 – No Authentication, Type 2 – Clear Text & Type 3 – Cryptographic (MD5 or SHA).
- OSPF Authentication can be configured on a per area or per interface on OSPF Protocols.
- But any OSPF authentication passwords has to be configured on the interface only.
- OSPF authentication type 1 means no authentication and is the default on any link.
- OSPF Authentication type 2 First enable authentication & create password on interface.

Authentication on RIP:

- Routing Information Protocol Version 1 (RIPv1) doesn’t support authentication.
- Routing Information Protocol Version 2 (RIPv2) does support authentication.
- RIP authentication is one of the protocol enhancements that appeared in Version 2.
- RIPv2 allows packets to be authenticated plain text password or secure MD5 hash.
- Authenticatio ensure that unauthorized equipment cannot affect how traffic is routed.
- Authentication in RIP comes in two variations, the plain text mode and the MD5 mode.



Initial Routers Configuration
R1(config)# interface ethernet 0/0 R1(config-if)# ip address 192.168.1.1 255.255.255.0 R1(config-if)# no shutdown R1(config-if)# exit R1(config)# interface loopback 1 R1(config-if)# ip add 1.1.1.1 255.0.0.0 R1(config-if)# no shutdown
R2(config)# interface ethernet 0/0 R2(config-if)# ip add 192.168.1.2 255.255.255.0 R2(config-if)# no shutdown R2(config-if)# exit R2(config)# interface loopback 2 R2(config-if)# ip address 2.2.2.2 255.0.0.0 R2(config-if)# no shutdown

RIP Authentication:

RIP Configuration
R1(config)#router rip R1(config-router)#network 192.168.1.0 R1(config-router)#network 1.0.0.0 R1(config-router)#no auto-summary R1(config-router)#version 2
R2(config)#router rip R2(config-router)#network 192.168.1.0 R2(config-router)#network 2.0.0.0 R2(config-router)#no auto-summary R2(config-router)#version 2
R1#show ip route rip R1#show ip route

RIP Protocols Authentication
R1(config)#key chain cisco R1(config-keychain)#key 1 R1(config-keychain-key)#key-string cisco1 R1(config-keychain-key)#exit R1(config-keychain)#exit R1(config)# interface ethernet 0/0 R1(config-if)#ip rip authentication mode md5 R1(config-if)#ip rip authentication key-chain cisco
R2(config)#key chain cisco R2(config-keychain)#key 1 R2(config-keychain-key)#key-string cisco1 R2(config-keychain-key)#exit R2(config-keychain)#exit R2(config)# interface ethernet 0/0 R2(config-if)#ip rip authentication mode md5 R2(config-if)#ip rip authentication key-chain cisco

EIGRP Authentication:

EIGRP Configuration
R1(config)#router eigrp 100 R1(config-router)#network 192.168.1.0 R1(config-router)#network 1.0.0.0 R1(config-router)#no auto-summary
R2(config)#router eigrp 100 R2(config-router)#network 192.168.1.0 R2(config-router)#network 2.0.0.0 R2(config-router)#no auto-summary
R1#show ip route eigrp
R2#show ip route eigrp

EIGRP Protocols Authentication
R1(config)#key chain cisco R1(config-keychain)#key 1 R1(config-keychain-key)#key-string cisco1 R1(config-keychain-key)#exit R1(config-keychain)#exit R1(config)# interface ethernet 0/0 R1(config-if)# ip authentication mode eigrp 100 md5 R1(config-if)# ip authentication key-chain eigrp 100 cisco
R2(config)#key chain cisco R2(config-keychain)#key 1 R2(config-keychain-key)#key-string cisco1 R2(config-keychain-key)#exit R2(config-keychain)#exit R2(config)# interface ethernet 0/0 R2(config-if)# ip authentication mode eigrp 100 md5 R2(config-if)# ip authentication key-chain eigrp 100 cisco

OSPF Authentication:

OSPF Configuration
R1(config)#router ospf 1 R1(config-router)# network 192.168.1.0 0.0.0.255 area 0 R1(config-router)# network 1.0.0.0 0.255.255.255 area 0
R2(config)#router ospf 1 R2(config-router)# network 192.168.1.0 0.0.0.255 area 0 R2(config-router)# network 2.0.0.0 0.255.255.255 area 0
R1#show ip route ospf R1# show ip protocol

OSPF Protocols Authentication
R1(config)#key chain cisco R1(config-keychain)#key 1 R1(config-keychain-key)#key-string cisco1 R1(config-keychain-key)#exit R1(config-keychain)#exit R1(config)# interface ethernet 0/0 R1(config-if)#ip ospf authentication message-digest R1(config-if)#ip ospf authentication-key cisco1
R2(config)#key chain cisco R2(config-keychain)#key 1 R2(config-keychain-key)#key-string cisco1 R2(config-keychain-key)#exit R2(config-keychain)#exit R2(config)# interface ethernet 0/0 R2(config-if)#ip ospf authentication message-digest R2(config-if)#ip ospf authentication-key cisco1

OSPF Plain Text Authentication
R1(config)# interface ethernet 0/0 R1(config-if)#ip ospf authentication R1(config-if)#ip ospf authentication-key 123
R2(config)# interface ethernet 0/0 R2(config-if)#ip ospf authentication R2(config-if)#ip ospf authentication-key 123
R1# show ip ospf interface Ethernet 0/0 R1# debug ip ospf packet
R2# show ip ospf interface Ethernet 0/0 R2# debug ip ospf packet

OSPF MD5 Authentication
R1(config)# interface ethernet 0/0 R1(config-if)#ip ospf message-digest-key 1 md5 123 R1(config-if)#ip ospf authentication message-digest
R2(config)# interface ethernet 0/0 R2(config-if)#ip ospf message-digest-key 1 md5 123 R2(config-if)#ip ospf authentication message-digest
R1# show ip ospf interface Ethernet 0/0 R1# debug ip ospf packet
R2# show ip ospf interface Ethernet 0/0 R2# debug ip ospf packet

```
*Aug 31 14:16:22.563: OSPF-1 PAK  : rcv. v:2 t:5 l:84 rid:1.1.1.1 aid:0.0.0.0 chk:821A aut:1 auk: from Ethernet0/0
*Aug 31 14:16:23.610: OSPF-1 PAK  : rcv. v:2 t:4 l:64 rid:1.1.1.1 aid:0.0.0.0 chk:B101 aut:1 auk: from Ethernet0/0
R2#
*Aug 31 14:16:28.168: OSPF-1 PAK  : rcv. v:2 t:1 l:48 rid:1.1.1.1 aid:0.0.0.0 chk:E48F aut:1 auk: from Ethernet0/0
R2#
*Aug 31 14:16:37.240: OSPF-1 PAK  : rcv. v:2 t:1 l:48 rid:1.1.1.1 aid:0.0.0.0 chk:E48F aut:1 auk: from Ethernet0/0
R2#
*Aug 31 14:16:47.063: OSPF-1 PAK  : rcv. v:2 t:1 l:48 rid:1.1.1.1 aid:0.0.0.0 chk:E48F aut:1 auk: from Ethernet0/0
R2#
```

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R2#
*Aug 31 14:19:10.301: OSPF-1 PAK  : rcv. v:2 t:1 l:48 rid:1.1.1.1 aid:0.0.0.0 chk:0 aut:2 keyid:1
ernet0/0
R2#
*Aug 31 14:19:19.415: OSPF-1 PAK  : rcv. v:2 t:1 l:48 rid:1.1.1.1 aid:0.0.0.0 chk:0 aut:2 keyid:1
ernet0/0
R2#
*Aug 31 14:19:28.947: OSPF-1 PAK  : rcv. v:2 t:1 l:48 rid:1.1.1.1 aid:0.0.0.0 chk:0 aut:2 keyid:1
ernet0/0
```